

EE 372 Assignment 2: Performance Analysis of M-PSK Modulation

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1 Introduction

In this assignment, the performance of M -ary Phase Shift Keying (M-PSK) modulation in the presence of Additive White Gaussian Noise (AWGN) is examined. The simulations are implemented using the Python programming language, utilizing libraries such as `numpy` for signal processing, `matplotlib.pyplot` for visualization, and `scipy.special` for theoretical error function calculations [2]. By utilizing Monte Carlo simulations with $N = 10^5$ trials, the Symbol Error Rate (SER) for $M \in \{8, 16\}$ is verified and the impact of bit-to-symbol mapping (Binary vs. Gray) on the Bit Error Rate (BER) is evaluated using transition probability matrices.

2 Part A: Analysis of Symbol Error Rate (SER)

The receiver's task is modeled as a Multiple Hypothesis Testing problem where the detector chooses the hypothesis $\hat{H}$ that minimizes the Euclidean distance to the received signal [1]. As M increases, the decision regions in the complex plane become narrower, making the system more sensitive to noise variance [1].

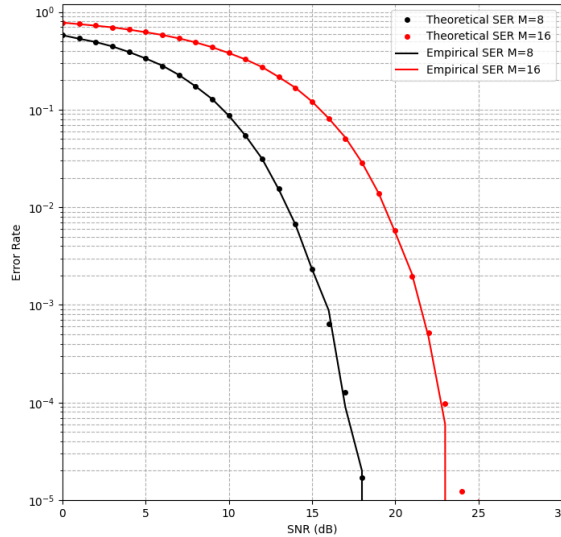


Figure 1: Empirical SER vs. theoretical $P_s = \text{erfc}(\sqrt{\gamma_s} \sin(\pi/M))$ for 8-PSK and 16-PSK.

2.1 Discussion on SER Behavior (M and SNR)

The results illustrate the power penalty associated with increasing spectral efficiency. To maintain a constant SER, 16-PSK requires significantly higher SNR compared to 8-PSK due to its reduced noise margin between adjacent constellation points [2].

Here it can be said that the empirical lines align almost perfectly with the theoretical markers, demonstrates that our relative frequency estimator is consistent and unbiased (Chapter 7) [1]. The drop-off observed at high SNR values (e.g., above 20 dB for $M=8$) is a result of the finite sample size ($N = 10^5$). At these levels, the probability of error becomes so low that rare events are no longer captured. This shows the practical limits of Monte Carlo simulation.

3 Part B: Bit Error Rate and Mapping Analysis

In this section, we analyze how symbol errors translate into bit errors. We utilize simply the rotational symmetry of the PSK constellation to estimate the transition probability matrix P_{ij} by simulating the transmission of a single symbol (s_0).

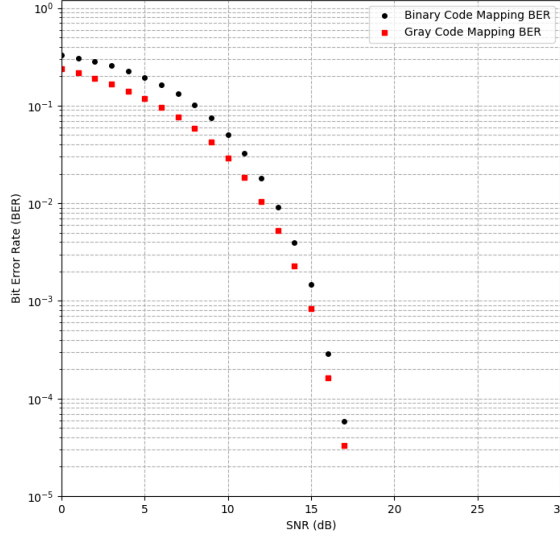


Figure 2: BER Comparison for 8-PSK: Binary vs. Gray Mapping Analysis.

3.1 Analysis of Gray Coding Gain

The comparison reveals that while the SER remains constant regardless of mapping, the BER is heavily dependent on the bit assignment. Gray mapping ensures that adjacent symbol errors result in only a single bit flip ($d_{ij} = 1$), whereas Binary mapping may result in multiple bit errors (up to k bits) for the same symbol mistake [1]. At high SNR, the Gray BER converges to approximately $SER/\log_2(M)$, proving that the most likely errors occur between nearest neighbor constellation points.

4 Transition Probability Verification

Utilizing the estimated transition matrix P_{ij} and the Hamming distance matrix D_{ij} , the expected BER is calculated as $E[BER] = \frac{1}{kM} \sum \sum P_{ij} D_{ij}$. (The assignment PDF gives the formula without $1/kM$ but [1] trivially takes $1/kM$ into account.) Also at high SNR, for Gray mapping, the SER/BER ratio is observed to be approximately $k = 3$. This is because almost all symbol errors are transitions to immediate neighbors, which Gray coding restricts to a single bit error. Here I tried to explain it via a table:

Mapping Type	Symbol Error Impact	High SNR Relation	Bit Reliability
Binary Index	High (Multiple Flips)	$BER \approx SER$ (Worst case)	Lower
Gray Code	Low (Single Flip)	$BER \approx SER/3$	Higher

Table 1: Analysis of mapping strategies on 8-PSK bit error probability.

5 Conclusion

This study confirms that M -PSK performance in AWGN is governed by both signal geometry and statistical mapping. While the SER is fixed by the decision boundaries and noise variance, the BER can be significantly optimized through Gray mapping [2]. Our results validate that the empirical transition probabilities derived from Monte Carlo simulations accurately predict analytical performance. [2].

References

- [1] Yates, R. D., & Goodman, D. J. *Probability and Stochastic Processes: A Friendly Introduction for Electrical and Computer Engineers*. Wiley.
- [2] Viswanathan, M. , *Digital Modulations using Python*